Machine Learning and Deep Learning
Lecture-06

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Spring 2026

Overview

- Support Vector Machines (SVM)
- Performance Metrices

Support Vector Machine (SVM)

Support Vector Machine (SVM)

- SVMs are supervised learning algorithms used for classification and regression tasks.
- SVMs find optimal decision boundaries (hyperplanes) that maximize the margin between classes, which improves generalization and reduces overfitting.
- They can handle non-linear problems by mapping data to high-dimensional feature spaces using kernel functions.
- SVMs are trained through convex optimization, ensuring convergence to a global optimum, and are grounded in statistical learning theory.

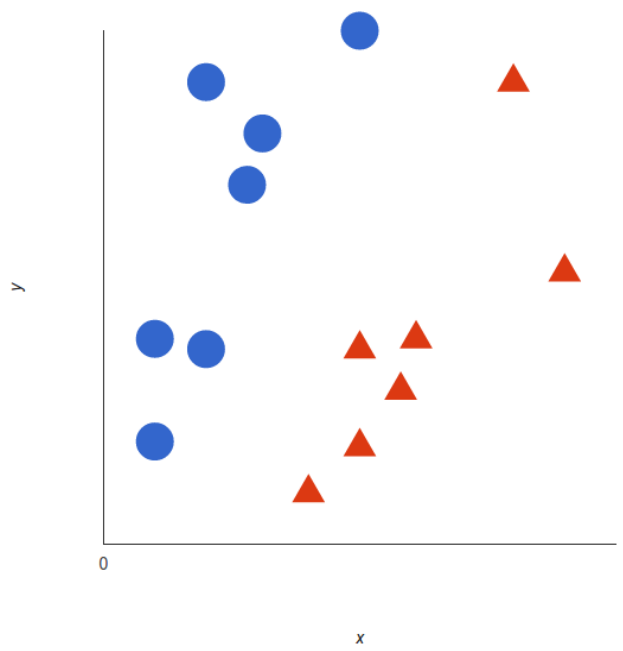
Classification and Regression

- Classification is a supervised learning task that predicts discrete categorical labels for new data instances based on their features.
- Regression is a supervised learning task that predicts continuous numerical values and can be either linear or non-linear.
- Regression models use loss functions to quantify prediction error by measuring the distance between predicted and actual values.
- Common loss functions include Mean Squared Error (MSE) for general cases and Huber loss for robustness against outliers.

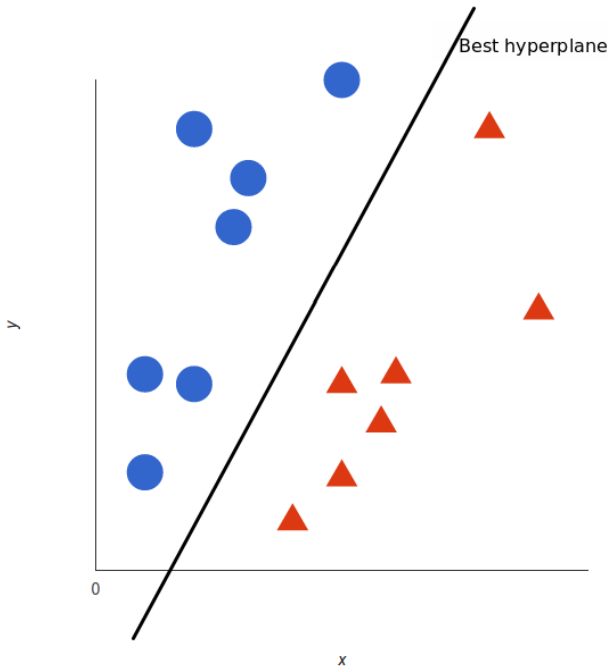
Classification and Regression

- Non-linear SVMs use kernel functions to implicitly map data into higher-dimensional feature spaces where linear separation becomes possible.
- The kernel trick enables learning non-linear decision boundaries without explicitly computing high-dimensional transformations, reducing computational cost.
- Common kernels include polynomial, radial basis function (RBF), and sigmoid kernels, each suited to different data patterns.
- SVMs have demonstrated strong performance in applications like handwriting recognition, face detection, and text classification, often matching or exceeding deep learning methods with smaller training datasets.

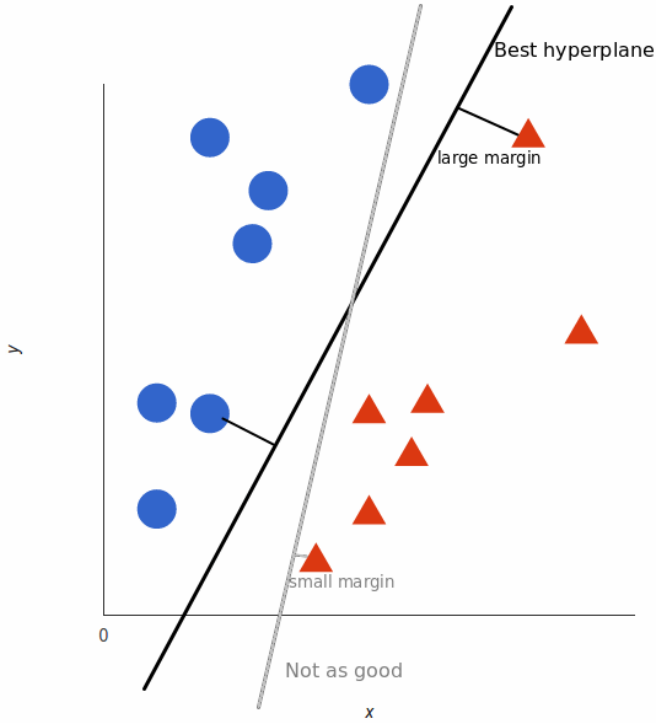
Support Vector Machine (SVM)



- Data: *red* and *blue*, data has two features: x and *y*.
- We want a classifier that, given a pair of *(x,y)* coordinates, outputs if it's either *red* or *blue*.



- SVM takes data points and outputs hyperplane (In 2D it's simply a line (decision boundary)) that separates tags.

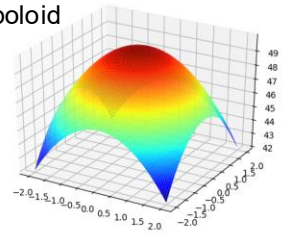


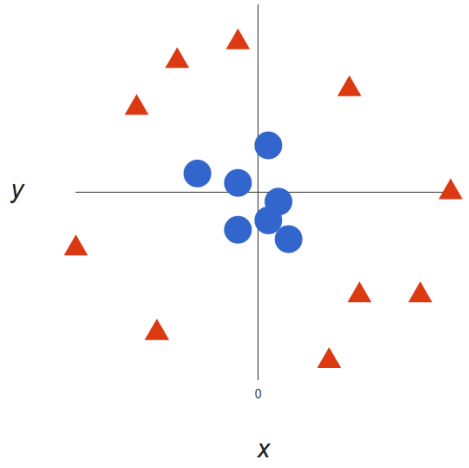
- SVM maximizes margins from both tags.
- Not all hyperplanes are created equal.

Img src: SVM By Bruno Stecanella

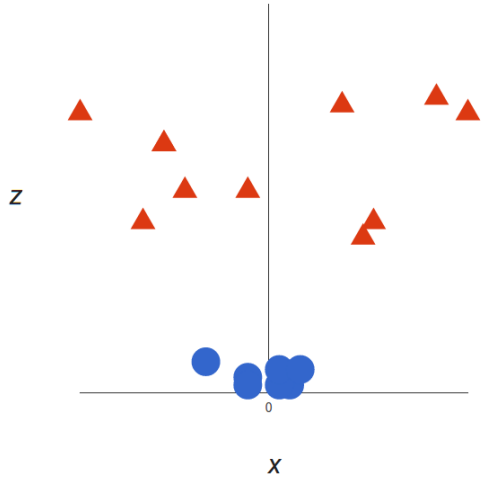
SVM: Nonlinear Data

Circular Paraboloid

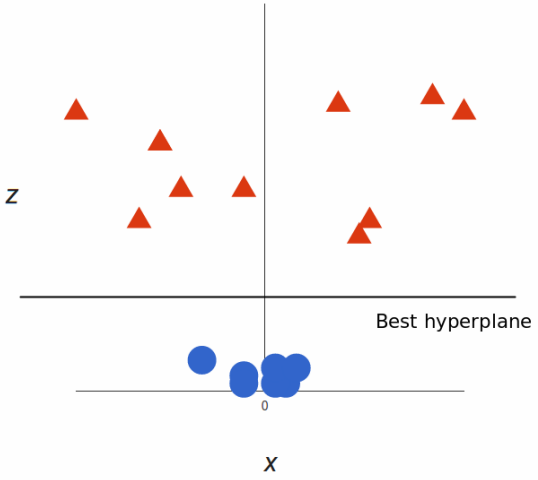




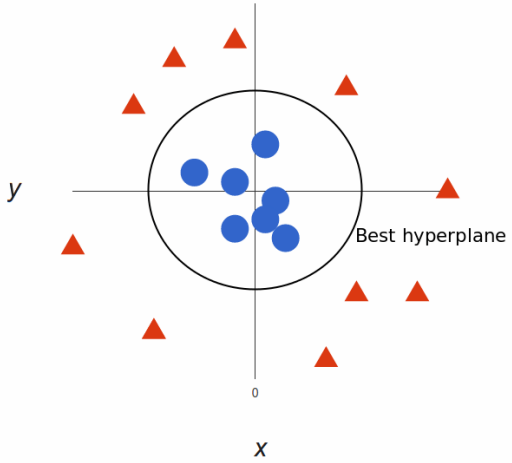
- Add z dimension
- Rule: $z = x^2 + y^2$ (Circular Paraboloid)



- Data in two linearly separated groups



- In 3D, hyperplane is a plane parallel to x axis at a certain z (z = 1).



- Decision boundary is a circumference of radius 1 (separates both tags using SVM)

- Shows how kernels help handle nonlinear boundaries.
- Many disease patterns are nonlinear (e.g., heart disease risk surfaces).
- This illustration of mapping to higher dimension explains kernel intuition elegantly.

SVM: Parameter C

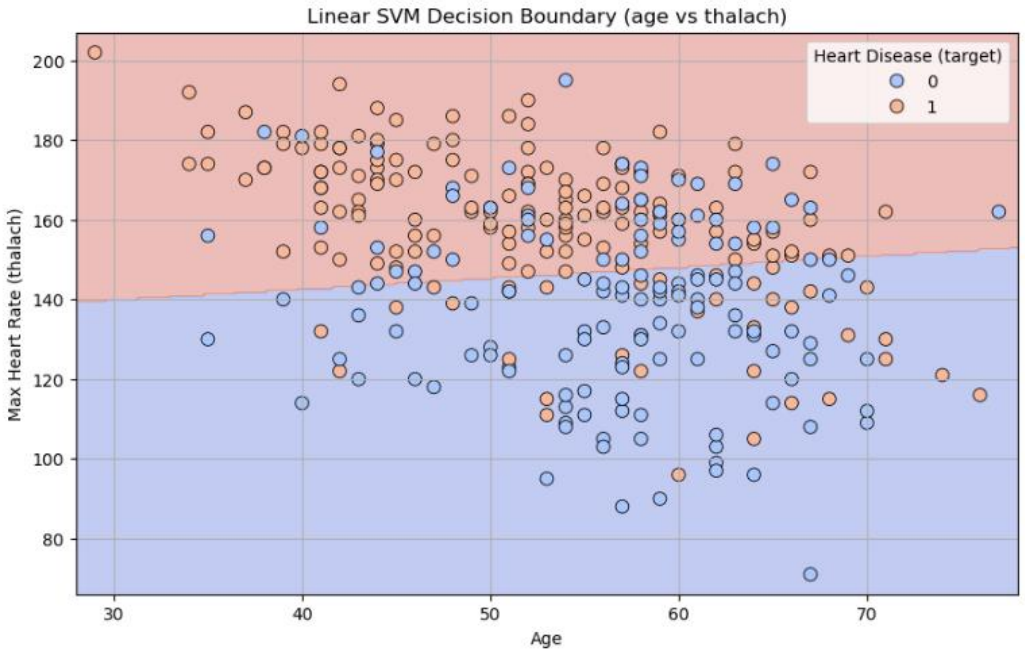
- C is a regularization parameter that controls the trade-off between maximizing the margin and minimizing classification errors on training data.
- Large C (e.g., 100, 1000): Prioritizes correct classification of all training points, resulting in narrow margins and potential overfitting.
- Small C (e.g., 0.01, 0.1): Tolerates some misclassifications to achieve wider margins, improving generalization to unseen data.
- For RBF kernel SVMs, two critical hyperparameters must be tuned together: C (regularization) and γ (gamma), which controls kernel width and decision boundary complexity.
- Hyperparameter tuning is performed using grid search with cross-validation or more efficient methods like randomized search or Bayesian optimization.

SVM: Kernel

- SVMs use kernel functions to measure similarity between data points and enable flexible decision boundaries.
  - Linear kernel: Creates linear decision boundaries, efficient for high-dimensional data (text classification).
  - Non-linear kernels (RBF, polynomial): Map data into higher-dimensional spaces where linear separation becomes possible, producing non-linear boundaries in the original space.
- The kernel trick computes similarity (inner products) in high-dimensional space without explicit transformation, making SVMs computationally efficient.
- SVMs find the optimal hyperplane by maximizing the margin
  - Margin the distance between the decision boundary and the nearest points (support vectors) from each class.

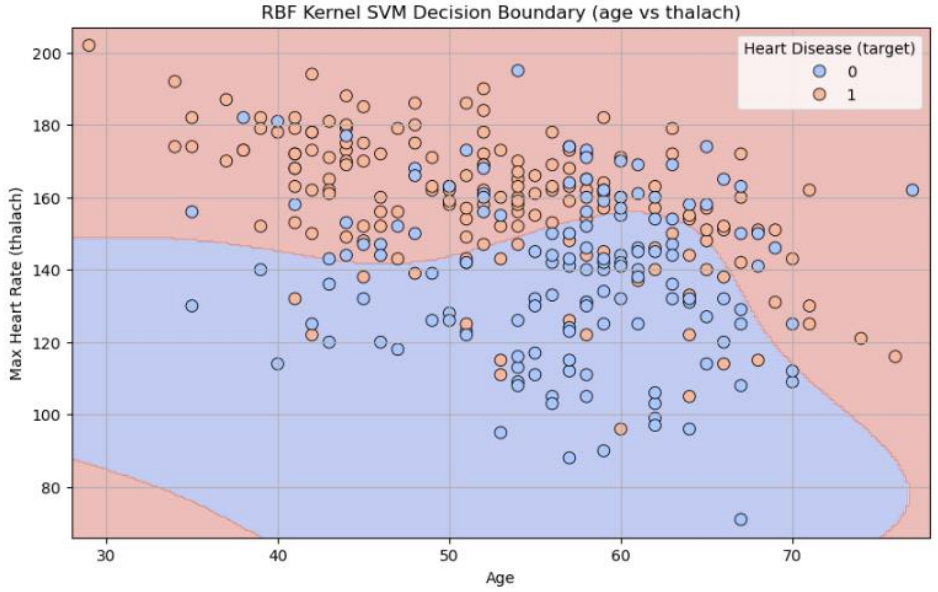
https://www.youtube.com/watch?v=_PwhiWxHK8o

Example



Linear SVM accuracy: 0.6973684210526315

- Decision boundary is a **straight line** (because kernel = linear).
- Shows how linear SVM separates patients based on age and heart rate.



RBF SVM accuracy: 0.6973684210526315

- Curved, nonlinear boundary.
- Adapts better to complex class distributions.

Support Vector Machine (SVM)

- Applications:
  - Handwriting recognition and face analysis, including pattern classification and regression-based tasks.
- Strengths:
  - Convex optimization guarantees a global optimum (no local minima).
  - Effective in high-dimensional spaces.
  - The trade-off between model complexity and training error can be explicitly controlled (via regularization).
- Weaknesses:
  - Performance and efficiency depend strongly on the choice of kernel and hyperparameters.
  - Training can be computationally expensive for large datasets.

Performance Metrics in Machine Learning

Classification Metrics

- Classification metrics evaluate a model's performance by measuring how well it predicts discrete class labels.
- Classification models produce discrete outputs (class labels or scores mapped to labels).
- Common classification metrics include:
  - Accuracy
  - Precision
  - Recall
  - F1 Score
  - ROC–AUC (Area Under the ROC Curve)

  [Lecture-02]
- The confusion matrix is a diagnostic tool, not a scalar metric, from which several metrics are derived.

AUC - ROC Curve

- A performance metric used to evaluate how well a classification model.
  - True Positive Rate (TPR) / Recall: How many of the actual positives did we catch?
  - False Positive Rate (FPR): How many negatives did we incorrectly flag as positive?
  - The curve shows the trade-off between sensitivity.
- A perfect model would go straight up the y-axis and then across the x-axis.

- Area Under the Curve (AUC): 2D area underneath that ROC plot.
- Provides a single number (between 0 and 1) to summarize the model's performance.
- AUC Score:
  - Perfect Model (1.0): Perfectly separates the two classes.
  - Great (0.8 – 0.9): Model has high discriminative power.
  - Fair (0.7): okay, but there's a fair amount of overlap/error.
  - 0.5 --> No Better Than Random
  - < 0.5: Recalibrate: Your model is actually predicting the opposite of the truth.

AUC - ROC Curve

```python
import numpy as np
from sklearn.metrics import accuracy_score, confusion_matrix, roc_auc_score, roc_curve

n = 10000
ratio = .95
n_0 = int((1-ratio) * n)
n_1 = int(ratio * n)

y = np.array([0] * n_0 + [1] * n_1)
# below are the probabilities obtained from a hypothetical model that always predicts the majority class
# probability of predicting class 1 is going to be 100%
y_proba = np.array([1]*n)
y_pred = y_proba > .5
```

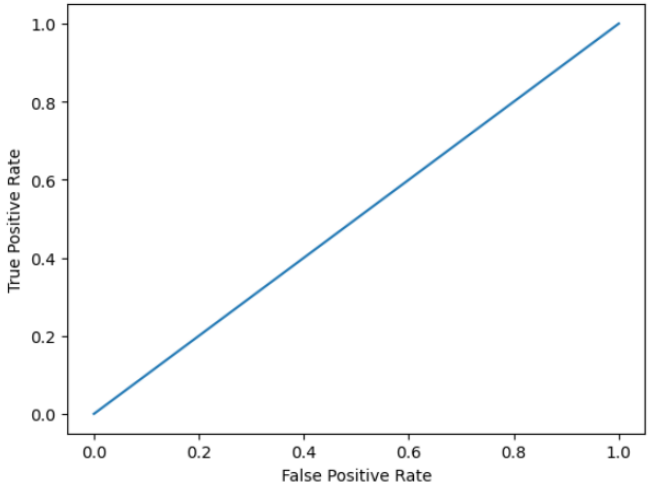
```python
# below are the probabilities obtained from a hypothetical model that doesn't always predict the mode
y_proba_2 = np.array(
    np.random.uniform(0, .7, n_0).tolist() +
    np.random.uniform(.3, 1, n_1).tolist()
)
y_pred_2 = y_proba_2 > .5
```

```python
import matplotlib.pyplot as plt

def plot_roc_curve(true_y, y_prob):
    """
    plots the roc curve based of the probabilities
    """

    fpr, tpr, thresholds = roc_curve(true_y, y_prob)
    plt.plot(fpr, tpr)
    plt.xlabel('False Positive Rate')
    plt.ylabel('True Positive Rate')
```

AUC - ROC Curve

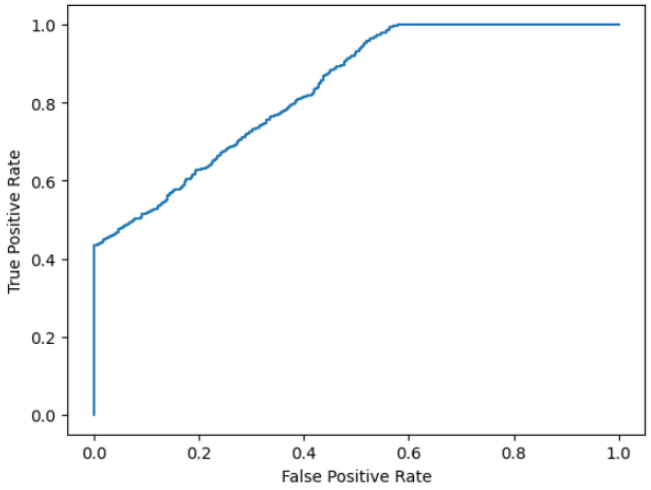
```
plot_roc_curve(y, y_proba)
print(f'model 1 AUC score: {roc_auc_score(y, y_proba)}')
```

```
model 1 AUC score: 0.5
```




```
plot_roc_curve(y, y_proba_2)
print(f'model 2 AUC score: {roc_auc_score(y, y_proba_2)}')
```

```
model 2 AUC score: 0.8334048421052631
```



Regression Metrics

- Regression models produce continuous-valued outputs.
- Their performance is evaluated by computing a distance or deviation between the predicted values and the ground-truth targets.
- Common regression evaluation metrics include:
  - Mean Absolute Error (MAE)
  - Mean Squared Error (MSE)
  - Root Mean Squared Error (RMSE)
  - R^2 (Coefficient of Determination)

Good to know!!

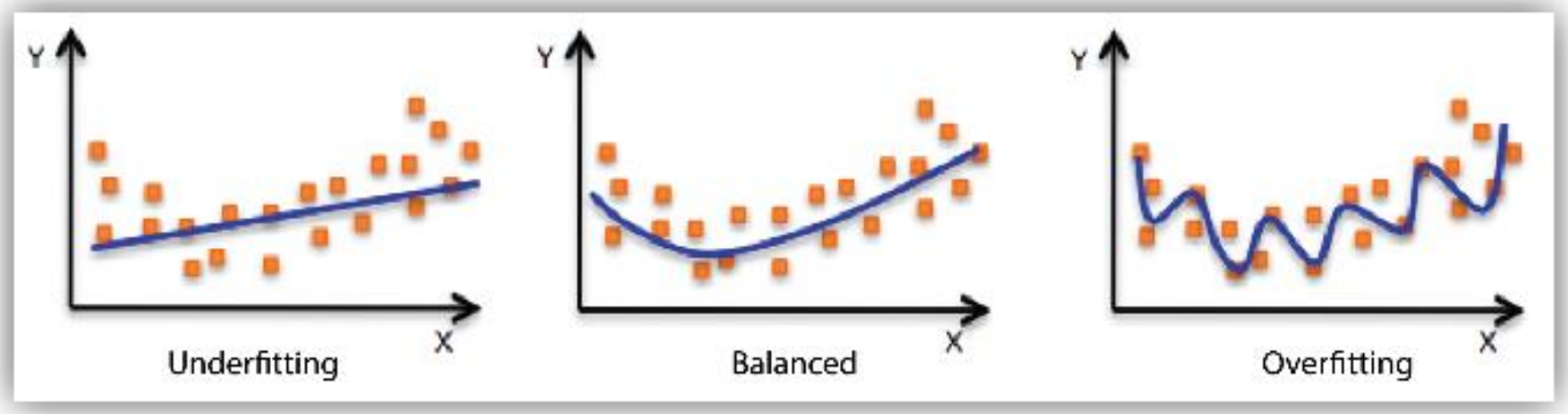
Model Fit

- Model Fit --> Improving Model Accuracy

- Model performance can be improved by:
  - Collecting more training data
  - Feature processing, such as adding relevant variables and improving feature representations
  - Model parameter tuning (hyperparameter optimization)

- Underfitting: A model performs poorly on training data because it is too simple to capture the relationship between the input features X and target values y.

- Overfitting: A model performs well on training data but poorly on test data, indicating that it has failed to generalize to unseen examples.



Case: High Accuracy in Imbalanced Data

```python
import numpy as np
from sklearn.metrics import accuracy_score, confusion_matrix, roc_auc_score, roc_curve

n = 10000
ratio = .95
n_0 = int((1-ratio) * n)
n_1 = int(ratio * n)

y = np.array([0] * n_0 + [1] * n_1)
# below are the probabilities obtained from a hypothetical model that always predicts the majority class
# probability of predicting class 1 is going to be 100%
y_proba = np.array([1]*n)
y_pred = y_proba > .5

print(f'accuracy score: {accuracy_score(y, y_pred)}')
cf_mat = confusion_matrix(y, y_pred)
print('Confusion matrix')
print(cf_mat)
print(f'class 0 accuracy: {cf_mat[0][0]/n_0}')
print(f'class 1 accuracy: {cf_mat[1][1]/n_1}')
```

```
accuracy score: 0.95
Confusion matrix
[[    0   500]
 [    0 9500]]
class 0 accuracy: 0.0
class 1 accuracy: 1.0
```

Imbalance: Majority of values are one

Summary

- SVMs are supervised learning models used for classification and regression
- Learn a maximum-margin decision boundary to improve generalization
- Use kernel functions to handle non-linearly separable data via implicit feature mapping
- Key hyperparameters include C (regularization) and γ (gamma) for RBF kernels
- Strengths: global optimum, effective in high-dimensional spaces
- Limitations: kernel and hyperparameter sensitivity, high training cost for large datasets

Summary

- Classification metrics: Accuracy, Precision, Recall, F1-score, ROC–AUC
- Confusion matrix is a diagnostic tool used to derive multiple metrics
- Regression metrics: MAE, MSE, RMSE, R^2
- Model fit depends on data quantity, feature quality, and parameter tuning
- Underfitting: model too simple, poor training performance
- Overfitting: good training performance, poor generalization
- Accuracy alone can be misleading for imbalanced datasets